A Hard-to-Beat Rule for Week-Ahead Traffic Forecasting

The calendar-adjusted seasonal baseline

Traffic Forecasting Project

June 2026

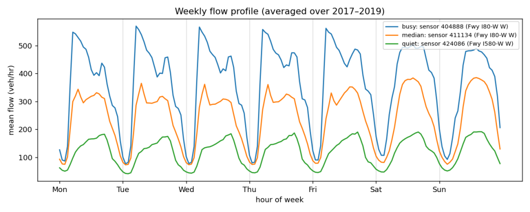
The problem

Forecast hourly highway **flow** (veh/h) **one week ahead** (168 hours) per sensor.

Key fact: traffic is **periodic with a one-week cycle** — the same weekday-and-hour repeats remarkably well.

Index by hour-of-week $\phi(t) = t \bmod 168$ (day-of-week $\times$ hour-of-day).

Data: Greater Bay Area, 2,352 sensors, hourly, 2017–2019 (LargeST).



Weekly flow profile: weekday commute double-peak, weekend hump.

The rule, in one line

Forecast = *(recency-weighted average of the same hour-of-week over the last few weeks)* $\times$ *(holiday factor on special days)*.

Three ingredients, building up:

1. A last-4-weeks hour-of-week **average** (the seasonal baseline)
2. **Recency weighting** (one parameter)
3. A **holiday/calendar factor** (known-ahead, multiplicative)

Reaches $R^2 = 0.949$, weekly-aggregate MAPE 3.55% — and we could not beat it.

Step 1 — last-4-weeks average (L4W)

Predict each hour by its recent average *at the same hour of week*:

$$b_t \;=\; \frac{y_{t-168} + y_{t-336} + y_{t-504} + y_{t-672}}{4}$$

(lags $= 1, 2, 3, 4$ weeks). Every term is $\geq$ one week old $\Rightarrow$ **leakage-free** for week-ahead.

Worked example — next Wednesday 08:00

Last four Wednesdays at 08:00: 540, 560, 535, 555 veh/h.

$$b = \tfrac{1}{4}(540 + 560 + 535 + 555) = \mathbf{547.5}\,\text{veh/h}$$

Averaging cancels week-to-week noise while staying recent.

Step 2 — recency weighting (EWMA)

If the level drifts, weight recent weeks more: weights $\propto \alpha^k$, $\alpha \approx 0.8$, i.e. about (0.34, 0.27, 0.22, 0.17) recent→old.

$$b_t^{\text{ewma}} = 0.34\, y_{t-168} + 0.27\, y_{t-336} + 0.22\, y_{t-504} + 0.17\, y_{t-672}$$

Worked example — a rising level

Last four same-hour values 500, 520, 540, 560 veh/h:

$$\text{flat} = 530 \qquad \text{vs} \qquad \text{EWMA} = \mathbf{537.4}\,\text{veh/h}$$

EWMA leans to the recent, higher values — tracking the drift the flat average lags.

The single decay α is the only "training" in the whole rule.

Step 3 — the holiday factor

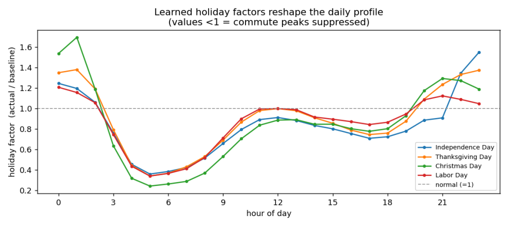
The baseline's blind spot: **holidays**. It averages four *normal* weeks and over-predicts (the commute vanishes).

8% of hours, but ∼23% of the error — and **known ahead**.

Fix: a multiplicative factor = *the average, over past occurrences of that holiday, of the ratio* $\frac{actual}{baseline}$ *at that hour.*

$$f = \tfrac{1}{2}(0.70 + 0.72) = 0.71$$

$$\hat{y} = b^{\text{ewma}} \times f$$



Learned factors by hour: mornings crushed to ∼0.4, late-evening > 1 (holiday-eve travel).

Putting it together — a holiday, hour by hour

July 4th at 17:00

baseline 490 veh/h $\times f$ 0.71 = **348** veh/h (a 29% correction)

hour	baseline (veh/h)	factor f	forecast (veh/h)
08:00	520	0.42	218
12:00	430	0.96	413
17:00	490	0.71	348
22:00	240	1.30	312

Same baseline, reshaped by the learned holiday profile.

Why it's hard to beat — it works

Model	R^2	RMSE (veh/h)	RMSE on holidays
Flat 4-week baseline	0.9404	40.14	66.96
+ recency (EWMA)	0.9435	39.10	65.99
+ holiday factor	**0.9493**	**37.01**	**47.71**

- Holiday factor cuts holiday-hour error **29%** ($67 \rightarrow 48$), ordinary hours untouched.
- Weekly-aggregate MAPE = **3.55**%.

Why it's hard to beat — it's at the ceiling

Best-possible *in-sample* fit on the test year itself (an upper bound):

Oracle (fit on the test year)	R^2
Static hour-of-week mean	0.938
+ oracle holiday effect	0.943
Our out-of-sample rule	**0.949**

Our *real* rule beats the *in-sample* static oracle — recency weighting tracks within-year drift a fixed mean cannot.

$\Rightarrow$ the remaining $\sim$5% is **irreducible incident noise**.

Why it's hard to beat — everything fancier failed

Attempt	Effect vs. the rule
Autoregression on residual (hourly)	$\leq +0.003\, R^2$, decays by a week out
Autoregression on weekly aggregate	$\leq +0.001\, R^2$
Ridge / XGBoost on calendar features	≈ 0 (or overfits)
Pooled panel AR (+ Fourier)	*worse*
Global gradient boosting	*worse*
Cross-regional common factor	exists, but *unforecastable*

A one- or two-parameter rule is at the ceiling: the signal is a deterministic calendar pattern; the rest is noise.

How to improve it

Not with fancier models of the *same* information — only with new information or a new objective:

1. **Known-ahead covariates** — scheduled events, planned closures (generalize the holiday factor; localized, help venue/corridor sensors).
2. **Longer horizons & more years** — multi-week, where a slow *level* governs error; matters from $\sim$3–4 weeks out.
3. **Distributional forecasts** — congestion risk, intervals; value in the tails, not the mean.
4. **Harder targets** — speed is far noisier ($R^2 \approx 0.76$).

Complexity should follow information and objective, not precede them.

For week-ahead flow, a calendar-adjusted seasonal average is not just a baseline — it is close to the best one can do.